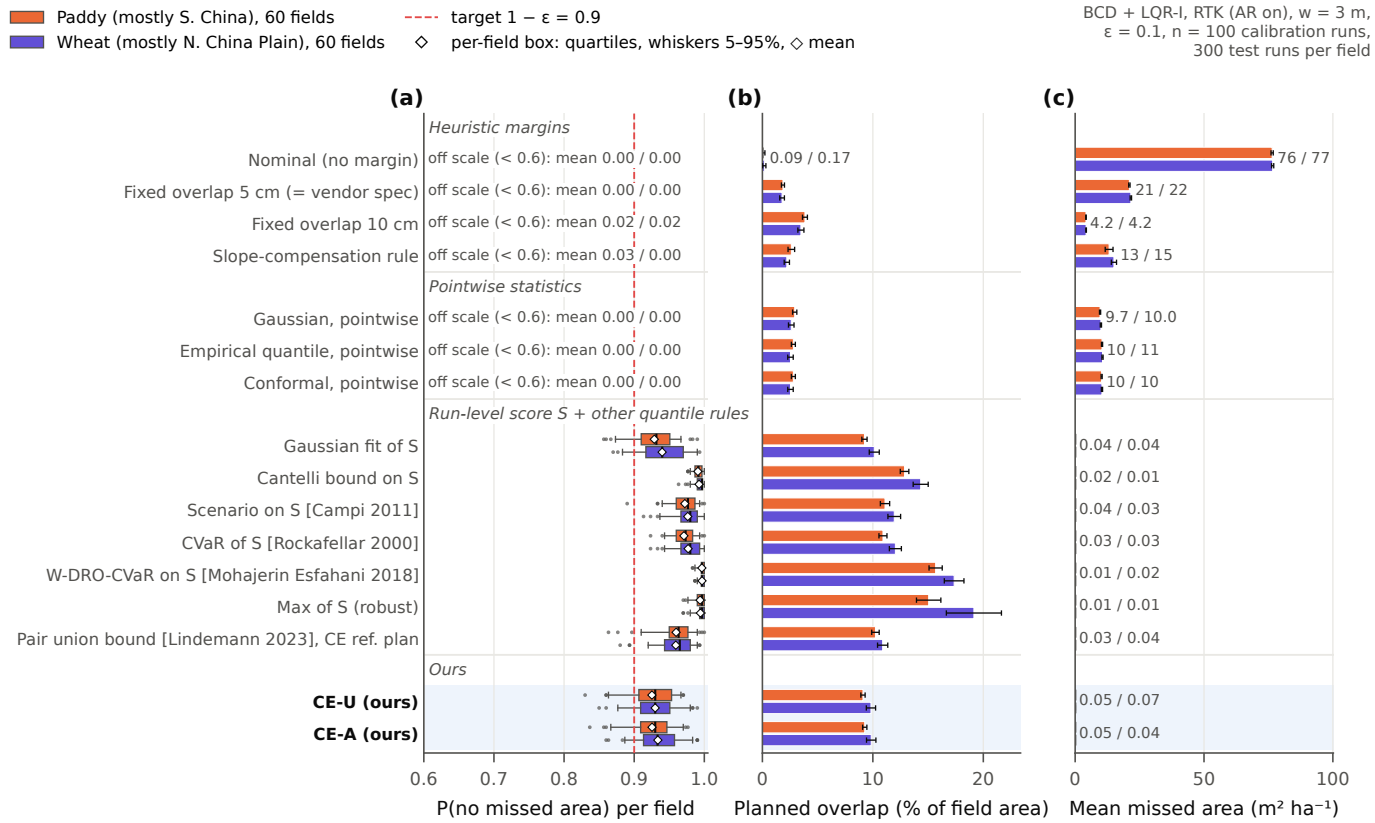
Paddy (mostly S. China), 60 fields
Wheat (mostly N. China Plain), 60 fields
target 1 − ε = 0.9
per-field box: quartiles, whiskers 5–95%, ◇ mean
BCD + LQR-I, RTK (AR on), w = 3 m,
ε = 0.1, n = 100 calibration runs,
300 test runs per field
(a)
(b)
(c)
Heuristic margins
Nominal (no margin)
Fixed overlap 5 cm (= vendor spec)
Fixed overlap 10 cm
Slope-compensation rule
Pointwise statistics
Gaussian, pointwise
Empirical quantile, pointwise
Conformal, pointwise
Run-level score S + other quantile rules
Gaussian fit of S
Cantelli bound on S
Scenario on S [Campi 2011]
CVaR of S [Rockafellar 2000]
W-DRO-CVaR on S [Mohajerin Esfahani 2018]
Max of S (robust)
Pair union bound [Lindemann 2023], CE ref. plan
Ours
CE-U (ours)
CE-A (ours)
off scale (< 0.6): mean 0.00 / 0.00
off scale (< 0.6): mean 0.02 / 0.02
off scale (< 0.6): mean 0.03 / 0.00
0.09 / 0.17
76 / 77
21 / 22
4.2 / 4.2
13 / 15
9.7 / 10.0
10 / 11
10 / 10
0.04 / 0.04
0.02 / 0.01
0.04 / 0.03
0.03 / 0.03
0.01 / 0.02
0.01 / 0.01
0.03 / 0.04
0.05 / 0.07
0.05 / 0.04
0.6
0.7
0.8
0.9
1.0
0
10
20
0
50
100
P(no missed area) per field
Planned overlap (% of field area)
Mean missed area (m² ha⁻¹)